

Test 9

Subject: Computer Networks

Total Marks: 50 | Time: 120 minutes

Q1. Name one application-layer protocol that commonly uses UDP as its transport protocol. [1 marks | Bloom: Understand]

Q2. State one primary function of the Network Layer in the TCP/IP model. [1 marks | Bloom: Understand]

Q3. What is the main function of the network layer? [2 marks | Bloom: Remember]

Q4. Explain how port numbers facilitate multiplexing and demultiplexing at the Transport Layer. [2 marks | Bloom: Understand]

Q5. Describe two primary functions of the Network Layer in the TCP/IP model, providing a brief explanation for each. [4 marks | Bloom: Understand]

Q6. Analyze what happens if a router does not have a route for a destination [5 marks | Bloom: Analyze]

Q7. Evaluate whether TCP or UDP is more suitable for file transfer. Justify your answer. [5 marks | Bloom: Evaluate]

Q8. A company has been allocated the Class B network address 172.16.0.0/16. They need to divide this into at least 5 subnets, with each subnet accommodating a minimum of 500 hosts. Design the subnetting scheme. For the first two usable subnets, provide the following:

- a) Network Address
- b) Subnet Mask (in CIDR notation and dotted decimal)
- c) First Usable Host IP Address
- d) Last Usable Host IP Address
- e) Broadcast Address
- f) Maximum number of hosts supported by each subnet. [10 marks | Bloom: Apply]

Q9. A router has the following simplified routing table entries:

Destination Network | Next Hop | Interface

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192.168.10.0/24 | 10.0.0.1 | Eth0

192.168.20.0/24 | 10.0.0.2 | Eth1

192.168.30.0/24 | 10.0.0.3 | Eth2

0.0.0.0/0 (Default) | 10.0.0.4 | Eth3

For each of the following destination IP addresses, determine the Next Hop IP address and the outgoing Interface that the router will use to forward the packet. Justify your answer based on the

Q10. Describe the purpose of flow control in the Transport Layer and explain how TCP implements this using the sliding window mechanism. [10 marks | Bloom: Understand]